Setup Guide — please finish this BEFORE the tutorial

You need three things on your laptop:

Tool	What it is for
Python + NumPy + Matplotlib + Jupyter	Plotting graphs
VMD	Looking at molecules in 3D
PyMOL	Looking at molecules in 3D, making pretty pictures

If something does not work, do not panic — note down the error message and bring it to the tutorial. We will fix it together in the first 15 minutes.

Part A — Python and the plotting libraries

Step 1: Check whether Python is already installed

Open a terminal (**Windows:** search for "PowerShell"; **Mac:** search for "Terminal"; **Linux:** `Ctrl + Alt + T`) and type:

```
python3 --version
```

(On Windows, try `python --version` if `python3` does not work.)

If you see something like `Python 3.10.12` (any version **3.9 or newer**), go to Step 3.

Step 2: Install Python (only if Step 1 failed)

- **Windows:** download from https://www.python.org/downloads/ and run the installer. **Important: tick the box "Add python.exe to PATH" on the first screen.**
- **Mac:** download from https://www.python.org/downloads/ and run the installer.
- **Ubuntu / Linux:**

  ```
  sudo apt update
  sudo apt install python3 python3-pip
  ```

Close the terminal, open a new one, and repeat Step 1.

Step 3: Install the libraries

```
pip install numpy matplotlib jupyter
```

If `pip` is not found, use `pip3` or `python3 -m pip` instead:

```
python3 -m pip install numpy matplotlib jupyter
```

> **Ubuntu 23.04 or newer:** if you get an `externally-managed-environment` error, run this instead:
>
> ```
> sudo apt install python3-numpy python3-matplotlib jupyter-notebook
> ```

Step 4: Check that everything works

Download the tutorial folder, open a terminal **inside that folder**, and run:

```
python3 check_install.py
```

You should see `All good!` and a window with a test plot.

Step 5: Open the notebook

In the same terminal:

```
jupyter notebook
```

A page opens in your browser. Click on `1_plotting_tutorial.ipynb`.

Nothing works? Use Google Colab (no installation)

1. Go to https://colab.research.google.com and sign in with your Google account
2. `File → Upload notebook` → choose `1_plotting_tutorial.ipynb`
3. Click the folder icon on the left, make a folder called `data`, and upload `pendulum.csv` and `marks.csv` into it

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Part B — VMD

VMD is free, but you have to register on their website.

1. Go to https://www.ks.uiuc.edu/Development/Download/download.cgi?PackageName=VMD
2. Choose the latest version (1.9.4 or 2.0) for your operating system
3. Register with your name and email (takes one minute), accept the licence, and download

Install:

- **Windows:** run the downloaded `.msi` / `.exe` file. Start VMD from the Start menu.

- **Mac:** open the `.dmg` and drag VMD to Applications. The first time, right-click the app → `Open` (macOS blocks it otherwise). Apple Silicon (M1/M2/M3) Macs: choose the **ARM64** build.
- **Linux:** in the folder where the file was downloaded:

  ```
  tar -xzf vmd-*.tar.gz
  cd vmd-*/
  ./configure
  cd src
  sudo make install
  ```

  Then type `vmd` in a terminal.

Check: when VMD starts you should see three windows — *VMD Main*, a black *Display* window, and a console.

Part C — PyMOL

- **Ubuntu / Linux** (easiest):

  ```
  sudo apt install pymol
  ```

- **Windows / Mac:** download the installer from https://pymol.org/ and install it. It asks for a licence — students can get a free educational one at https://pymol.org/edu/
- **Alternative for any OS, if you use Anaconda / Miniconda:**

  ```
  conda install -c conda-forge pymol-open-source
  ```

Check: type `pymol` in a terminal (or open it from the Start menu / Applications). A window with a black viewer should open. In the command line at the top, type:

```
fetch 1ubq
```

If a protein appears, you are ready.